Conceptualization of an Integrated Procedure Model for Business Process Monitoring and Prediction

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***Abstract*—Predictive Process Monitoring (PPM) aims to improve operational business processes through the prediction of future behavior and process-related performance indicators. It includes a set of techniques to predict the future behavior of business processes based on knowledge of previously executed process instances. The need for rapid intelligent support in business process management brings growing attention to the field of PPM. A variety of machine learning (ML) based PPM techniques have been proposed by researchers to tackle several PPM-related prediction tasks, such as next activity, time, risk, or outcome. However, adopting such a PPM technique involves many complex tasks, including the selection and parametrization of a machine learning algorithm. Additionally, the novel domain lacks a set of guidelines on how to implement the methods to enable a broad and non-expert user group to benefit from it. As a remedy, this paper proposes a procedure model, which can serve as a step by step framework during the implementation and application of PPM. We conclude our work by demonstrating the applicability of our procedure model and by outlining our case study-based evaluation concept.**

***Index Terms*—prediction, monitoring, business process, procedure model**

I. Introduction

With the development of process mining techniques, we can collect and analyze executions of business processes through a variety of business process management (BPM) tools [1]. In traditional process mining, we take historical data of the business process as input, generate insights from it, and improve the business process according to the knowledge learned from the data [2]. However, the techniques in traditional process mining can only affect future instances. They are not able to detect and solve the problem at the moment that it occurs [3]. Predictive Process Monitoring (PPM) refers to a set of novel approaches that address this limitation. It analyzes historical event log data and the current execution of an ongoing instance in order to predict the future behavior of the running process instance [2]. PPM can be considered as an extension of process mining with predictive functionality. A business process instance is continuously monitored as long as the process is not finished. Therefore, PPM enables an organization to respond to potential issues or violations in real-time before they develop into more severe problems.

In 2012 the “Process Mining Manifesto” proposed several challenges of process mining, suggesting that the traditional offline analysis is not sufficient anymore to meet the requirements of online operational support [4]. Modern process mining should, therefore, be able to detect the case immediately when it deviates from the predefined path, be able to build a prediction model based on historical data to predict the expected behavior of the running case and be able to generate recommendations on future actions to achieve defined business benefits. These requirements can be satisfied with PPM. From a BPM perspective, PPM emancipates the labor of process management, which currently is still done by process managers, analysts, and engineers, based on their knowledge and experience. PPM provides methods to monitor running instances and predict future behavior. In the past seven years, a variety of PPM methods have been developed that address different prediction goals, such as time, next activity, outcome, or risk [2]. However, every method has its specific prediction goal, application domain, and requires distinct features from the input dataset. The adoption of such a PPM method involves many complex tasks, e.g., the selection and parametrization of a machine learning algorithm. As a consequence, PPM methods are rarely applied in practice today. Additionally, a lack of planning or guidance is often the reason why such projects get out of control or fail [5]. What is missing is a general guideline that a non-expert, i.e., practitioner, can follow to implement PPM methods successfully. Therefore, this paper proposes the first instantiation of an integrated procedure model, which describes the standard steps of implementing a PPM method, starting from scenario analysis until actual deployment.

II. Research Design

This paper follows the design-oriented approach by Österle et al. [6], which suggests a four-phase iterative process to conduct IS research. The phases are *analysis, design, evaluation,* and *diffusion* [6]. *Analysis* refers to problem and research objective identification, which is described in Section I. In this paper, we design the first concept of an integrated procedure model for PPM. Due to the lack of PPM-specific reference models, we base our *analysis* on successful applications in related domains, which are enriched with specific features of PPM. To achieve this, we performed a literature review about existing process models in PPM and its related domains before *designing* the artifact in Section IV. The third step is to *evaluate* the designed artifact. We realized an initial evaluation through a pilot application (see Section V) and describe our detailed evaluation plan as part of our future work (see Section VI). This paper represents the *diffusion* step.

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III. Analysis

As PPM is still a young research field, to date, there exists no reference process to guide PPM method implementation. To remedy this, we performed a literature review to identify the state of the art of process models in BPM, PPM, and its related domains. For the structured literature review, we follow the established framework of vom Brocke et al. [7]. As defined in the "Process Mining Manifesto" in 2012 [4], PPM can be seen as a child field of process mining, which is an interdisciplinary domain in between data mining and process analysis. Therefore, "data mining" is considered as one of the keywords to limit the subject areas. The final Boolean search string that we applied is "process model" AND ("data mining" OR "process monitoring" OR "process prediction").

The databases used in the literature review are Scopus and Web of Science. The initial search resulted in 1902 manuscripts. We filtered the findings based on the subject area ("Computer Science", "Engineering", "Mathematics", "Decision Sciences", and "Business Management and Accounting") and excluded workshop papers as well as papers under six pages, as these will not include a sufficiently substantial contribution. After removing duplicate entries, 1170 hits remained. By checking the titles and abstracts, we finally identified 25 publications relevant to our research, from which we identified an additional six papers through a backward search.

A. Related Work from the Business Process Management Domain

As mentioned above, the PPM domain currently still lacks comprehensive conceptual models. However, there are some works, which provide first insights and can serve as reference points for the implementation of PPM. Poll et al. [8] introduced a process forecasting reference process, which describes the essential steps to be followed in a PPM endeavor. Fig. 1 depicts the six consecutive steps of the model. Even though this reference process provides some guidance about the general steps to be undertaken when implementing PPM, it merely gives a general overview. Therefore, it better suits someone already familiar with the field. There are no sub-steps defined for the six main steps, which detail the activities and methods to be applied, nor is there any form of iteration or step backward included within the model, which would be common in data analysis and prediction approach (i.e., fast prototyping).

The research field of PPM is very active. A procedure model for PPM implementation should, therefore, build upon current research designs and incorporate the methods that are applied. A considerable amount of PPM methods has been developed in recent years. [9] provided a first overview of these methods. [10] recently presented an empirical comparison of several classification techniques for next event prediction. Similarly, [11] provide a systematic review and taxonomy of outcome-oriented predictive PPM methods, including an experimental evaluation. Another approach was taken towards the development of Nirdizati 2.0 [12], which is a web-based PPM engine for running business processes and analyzing them. All these works and the previously described reference process follow similar steps, which give a good first understanding of the elementary steps in PPM method implementation. However, they primarily focus on the technical components of a PPM implementation, especially on the data pre-processing and method selection, whereas there are other, e.g., business-related, perspectives to consider.

B. Related Work from the Data Mining Domain

The conceptual works from the Data Mining domain are much more extensive. In 2010 [13] compiled an overview of data mining and knowledge discovery process models (see Figure 2). Their figure shows that CRISP-DM (Cross-Industry Standard Process for Data Mining) [14] is a central process model, which integrated all previously existing approaches. For this reason, we focused on the analysis of the identified publications on works published after the year 2000. In the following, we briefly describe the identified models and their characteristics.

The earliest model among the results is the Cross-Industry Standard Process for Data Mining (CRISP-DM). CRISP-DM consists of 4 hierarchical levels (see Fig. 3) [14]: The first two levels list the general tasks covering all possible situations in a data mining project. The lower two levels are specific actions for different situations in each generic task. Figure 4 shows the major phases of CRISP-DM: The initial phase of the lifecycle *Business Understanding* includes objectives determination from both, the business and method perspectives, resource and constraint assessment, and project plan making. This phase is the decisive factor influencing the success of the project. The second phase *Data Understanding* collects initial data, gets insights into its basic characteristics, and examines data quality. Then, in the next phase *Data Preparation* the raw data is converted into a dataset that meets the project requirements in the format compatible with the modeling tools. The fourth phase *Modeling* builds the data mining model and evaluates the results from the method perspective. Finally, the evaluation for business objectives is done in the fifth phase *Evaluation*, the result of which decides whether to deploy or redo previous phases for the next step. If the result is positive, then the last phase is the *Deployment* of the solution in the respective business.

[15] have published a Six-step DMKD (data mining and knowledge discovery) Process Model applied in the healthcare industry in August 2000. They were interested in understanding the vast amounts of data generated in modern medicine every day. Although the process model is very similar to CRISP-DM, [15] put additional focus on examining the concept of knowledge discovery, which means converting the product into a piece of new information after executing a data mining process, as well as detailed preparation of the data. They claimed that "preparation of data is the key step on which the success of the entire knowledge discovery process depends". Compared to CRISP-DM, it introduced new feedback mechanisms among data mining, business understanding, data understanding, and data preparation sub-steps [15].

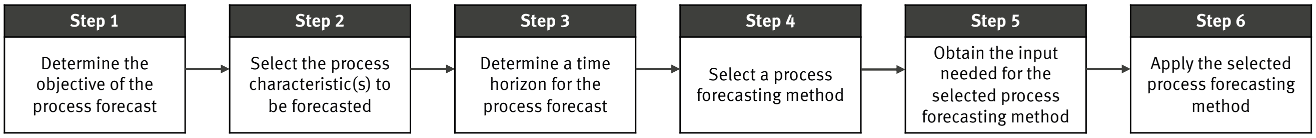

Fig. 1. Process Prediction Reference Process [8].

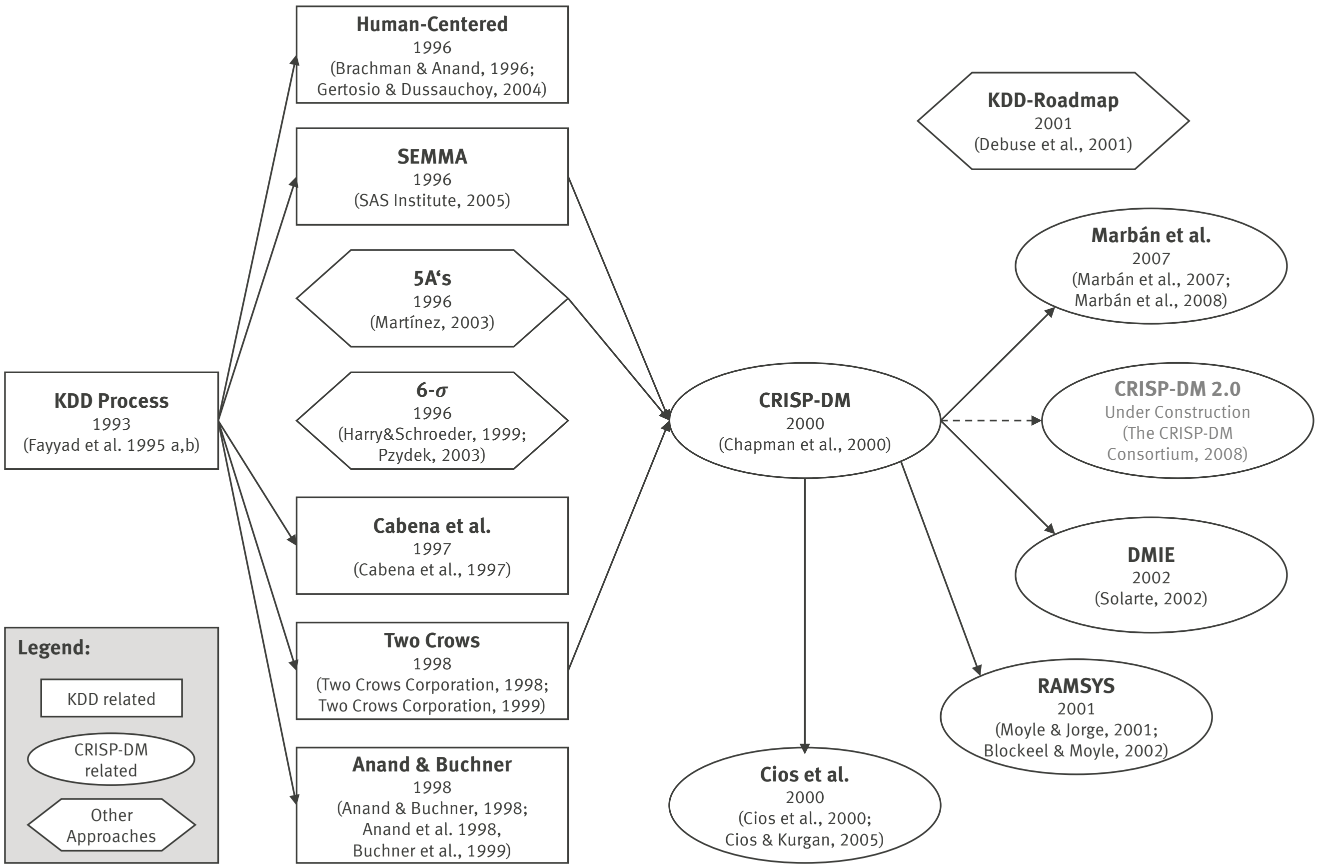

Fig. 2. Evolution of Data Mining and Knowledge Discovery Process Models [13]

RAMSYS, formally known as Rapid Collaborative Data Mining System, is a data mining methodology and system for groups geographically located in different places but working together on one data mining project collaboratively [16]. This methodology further refines CRISP-DM by adapting some of its generic tasks for collaborative work and by taking restrictions of working remotely into consideration.

The KDD (Knowledge Discovery in Databases, i.e., Data Mining) Roadmap proposes an 8-stage process with a feedback loop, which considers the entire process as a route, and each stage as a stop along the way [17]. Data Mining for Industrial Engineering (DMIE) is a methodology developed for implementing data mining techniques in industrial engineering in order to improve the effectiveness of solving problems. It provides a 5-step process model similar to CRISP-DM [18]. The Data Mining Engineering Process Model (DMEPM) was proposed to introduce engineering process-specific tasks into CRISP-DM. For the development, the engineering process models IEEE STD 1074 and ISO 12207 were combined and then enhanced with CRISP-DM tasks [19]. [20] emphasized the issues caused by clustering in a knowledge discovery project. Therefore, this paper adapted the CRISP-DM process model to incorporate clustering and introduced a domain expert role for the segmentation evaluation. [21] did a study in 2010 on which data mining procedure models were applied the most. CRISP-DM stood out, with 47% of projects using it. Following their survey, the authors integrated the various steps of the most popular models and called it the ”Refined Data Mining Process”. [22] proposed an advanced data preparation phase within CRISP-DM. They extend the phase by a third layer of subtasks, which includes 12 tasks and distinguishes between business, data, and methodological know-how.

Among the relevant findings of our literature review, four of the procedure models are adapted from CRISP-DM [15], [16],

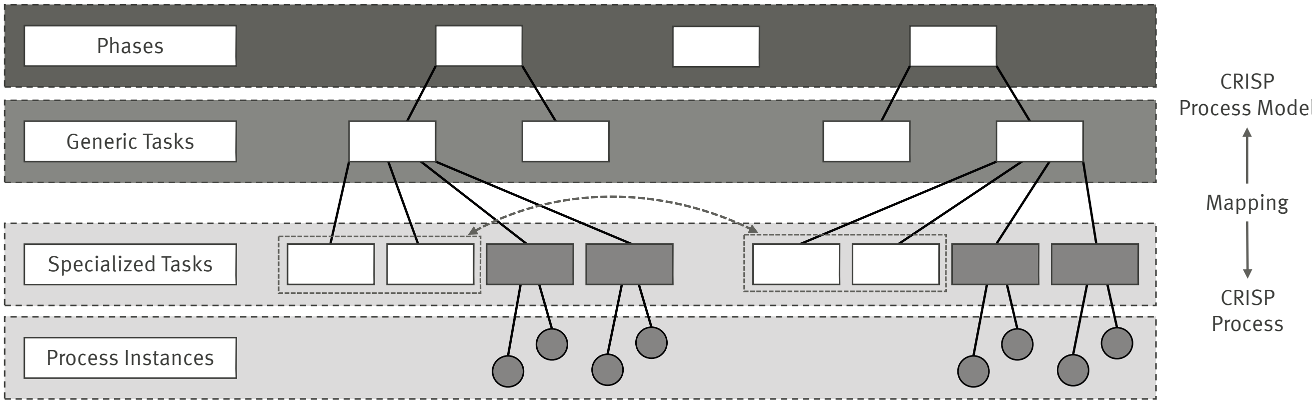

Fig. 3. CRISP-DM [14] Hiearchy Levels

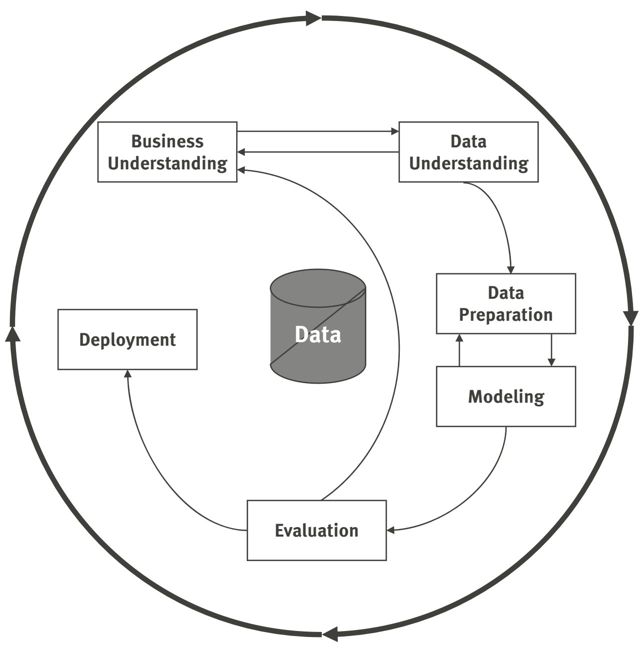

Fig. 4. Outline of CRISP-DM [14]

[19], [23], and 3 integrate several solutions alongside CRISP-DM [13], [24], [25]. This shows that besides its wide-spread application in practice [21], CRISP-DM also has a considerable influence on the development of new procedure models. Furthermore, it shows that it has a reasonable degree of generalization towards different kinds of data-driven projects. Due to its wide adoption and a high degree of generalization, we will adopt CRISP-DM as the basis for our PPM specific procedure model.

IV. Design

According to the result of the literature review, our Business Process Monitoring and Prediction Procedure Model (BPMP-PM) is designed based on CRISP-DM, adopting its structure and the project management related tasks, which are highly similar for most information system projects. BPMP-PM keeps the main phases Business Understanding, Data Understanding, Data Preparation, Evaluation, and Deployment. As introduced in Section I, the goal is not to develop an entirely new model but to adapt an appropriate model from a related domain as a starting point. For the adjustments, we take into account the related work from the PPM domain and our personal experience with PPM implementations.

The most notable changes are as follows: *Method Selection and Implementation* replace *Modeling* as the fourth phase (see Figure 5). Furthermore, we introduce the IS-based datasets into the model, which are required in any PPM project. The input data types of a data mining project are manifold, while the essential input of a PPM project is always at least one event log. Therefore, the *data* resource of CRISP-DM is replaced by PPMs *Event Logs*. Besides, a collection of existing PPM methods [9] is introduced to the model to support the user in the method selection step.

Although the general structure and also the project management related tasks are very similar, there are several detail tasks of BPMP-PM, which are different from CRISP-DM. Figure 6 shows an overview of the detail tasks in each phase of BPMP-PM. The details tasks and their differences to CRISP-DM will be elaborated further in this section.

A. Business Understanding

The first phase of a PPM implementation project is *Business Understanding*. Similar to most business projects, the major tasks in this phase are project background analysis, objective determination, situation assessment, and project plan creation. A PPM project is different from others in the third sub-step *Determine Technique Goal*. This step's primary purpose is determining which process to monitor and what to predict to achieve the business objectives. The required prediction type should be specified as well. According to a related literature review [9], the common prediction goals covered by the currently existing methods include time, categorical outcomes, sequence of outcomes/values, risk, inter-case metrics, and cost.

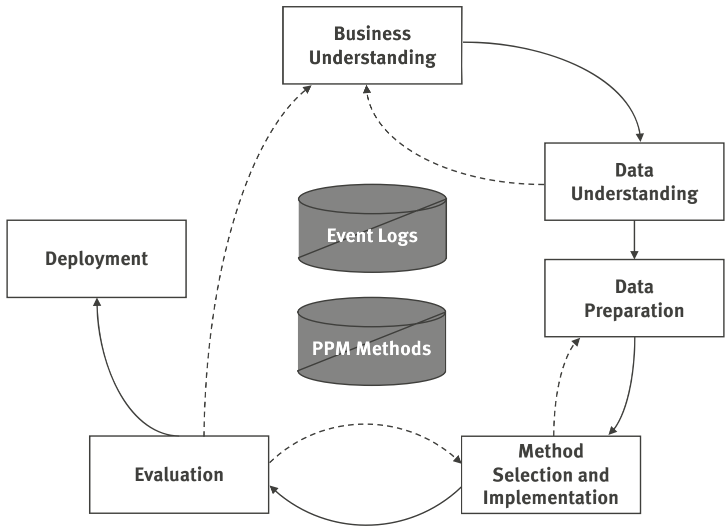

Fig. 5. Overview of the Business Process Monitoring and Prediction Procedure Model

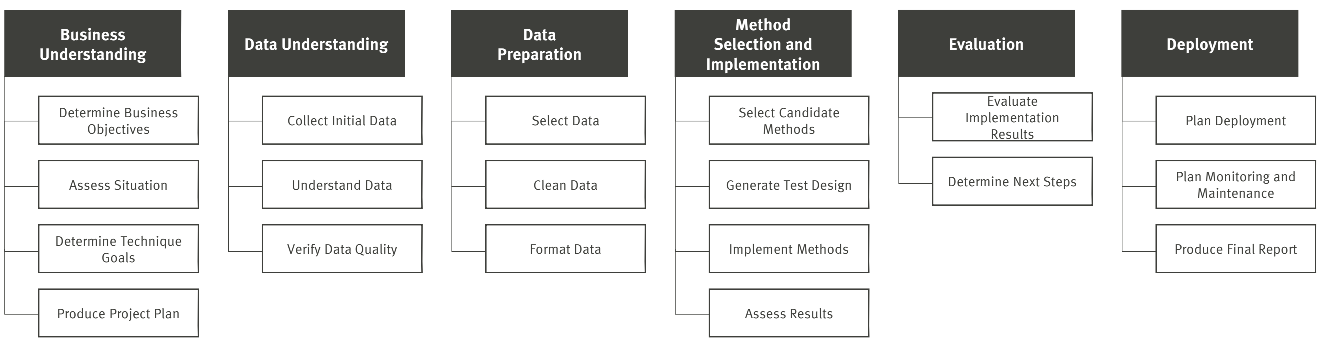

Fig. 6. Detail Tasks of the Business Process Monitoring and Prediction Procedure Model

B. Data Understanding

The next two phases deal with the event logs of the monitored process. The primary purpose of *Data Understanding* is to collect data and get familiar with it. The gained knowledge aids the subsequent *Data Preparation*. The first step is to collect log files from process management systems. A log file consists of multiple cases, while a case consists of multiple events. A record of each event at least contains a timestamp, the operation name (activity), and the ID for distinguishing between cases. With the ID and timestamp, the events belonging to the same case can be classified into one group and sorted by time, thereby learning what happened in each case [26]. It allows a better understanding of business processes. Moreover, the focus of PPM is different from regular data mining. It puts more emphasis on understanding traces rather than attributes. An individual event - apart from the entire trace - is meaningless. Therefore, the first thing to look at in the sub-step *Understand Data* is the basic statistics of the event logs, including the number of cases, events, activities, and timestamps. Moreover, different prediction types have their specific characteristics. For example, time prediction requires duration information, and sequence prediction emphasizes event occurrences. The second thing to focus on is the meaning of each attribute and corresponding values, especially for the attributes relevant to the prediction (i.e., the name of operations in next activity prediction). After that, the event logs can be visualized as a process model. A visual process model is the most intuitive way to understand a given business process. There are a lot of process mining tools available to analyze and visualize event logs, such as Celonis, Disco, Fujitsu, LANA, Minit, or ProM [27].

A situation, which may trigger stepping back to the first phase is when the knowledge learned in this step can improve the decisions in Business Understanding. The *Verify Data Quality* step detects quality problems in the dataset. There are six issues, which should be checked: spelling and format mistakes, invalid attribute values, invalid traces, missing values, outlier traces, and redundancy [28]. Among them, invalid traces and outlier traces are particularly relevant for PPM. Invalid traces refer to traces violating business and process rules. A rule could refer to the occurrence of an individual event: For example, the trace does not start with a previously specified start event. It can also analyze the relationship between events. For example, event A should be followed by event B, or event C should not occur if event D is not finished. An outlier trace refers to a trace, which rarely occurs in the event log. Analyzing the trace frequencies can be used to detect such instances. Most identified problems should be judged by business rules or expert knowledge. The process mining tools used in *Understand Data* can also help to detect data quality problems in this sub-step.

C. Data Preparation

The initial data collected in the last step are the event logs of the relevant business processes. A PPM project may also concern only a specific part of the available data. Thus, the first task in this step is to select the required data from the original event logs. Data selection is considered from three perspectives. One is that the project only focuses on certain kinds of instances. They can be selected through filtering attribute values. For example, in a credit card application censoring process, if the project is only interested in student applicants, the instances of student applicants can be selected by filtering an "Applicant Occupation" kind of attribute. Another perspective is that only a part of the process is involved in the project by defining the start and end events of the relevant sub-process. The last one is that the project requires certain kinds of events. For example, a project monitors order cancellations. Event logs about traces containing "Cancel Order" (or similar) events will be selected for the project. Fortunately, most process mining tools can filter traces so that they have to include a particular event. The scope of the monitored processes is defined in the *Determine Technique Goals* step. After that, the data problems discovered in *Verify Data Quality* can be solved. Generally, the treatments are to correct the values, ignore the problems if they do not influence the prediction, or remove the entire case. The particular situation for PPM is that a case is the basic unit of an event log. It is not allowed to keep a case while removing an individual event from the instance. One should either keep the event or remove the entire case. Usually, a domain expert will be consulted for this process. The previous operations performed on the dataset can be done in process mining tools. However, the tool used for the actual prediction depends on the selected PPM method. It is most likely different from the process mining tools used for *Data Understanding* and *Data Preparation*. Thus, the last step of Data Preparation is to export the prepared dataset into the format specified by the methods selected in the next step. Among existing PPM methods, the most popular tools are R, Python, and ProM [9]. The input format used by Python and R is usually a CSV file. Different from that, ProM usually uses the XES format, the de-facto standard for business process event logs [29].

D. Method Selection and Implementation

This step is entirely new compared to CRISP-DM. Different from a typical data mining project, this procedure model does not target the situation that the model, including the predictive method, has to be built entirely from scratch. According to an overview by Di Francescomarino et al., there currently exist at least 55 PPM-specific methods, originating from previous research efforts [9]. In a PPM implementation/application project, it usually is sufficient to select an appropriate method among existing ones instead of developing a new prediction method. Di Francescomarino et al. have structured the identified existing methods into a framework and categorized them by prediction type, input, tool, domain, and algorithm [9]. Figure 7 displays an excerpt of the framework focusing on the time prediction type. This framework is the primary reference point during method selection and depicted in Figure 5 as *PPM Methods* resource.

The available methods of the framework can be filtered from five perspectives. The first filter is the prediction type. Di Francescomarino et al. classified PPM methods into six types: time, categorical outcome, sequence of outcomes/values, risk, inter-case metrics, and cost. The prediction type of the project was already defined in *Determine Technique Goals*. Methods whose required inputs are not available in the project can be excluded. Another important filter is the tool availability. Only the implemented methods, where the implementations are available, can be considered further. Next, methods using similar algorithms can be compared. Often methods in one algorithm family have evolved based on the same original method. Analyzing them by algorithm group makes it easier to discover their distinctions and strengths. The methods matching the project requirements and dataset features will be selected. Moreover, some methods may have already compared their performance with other methods. The evaluation result helps to decide which method is more advanced and performs better. However, the number of methods to be selected for comparison in one algorithm group is not limited. If there are still too many methods left after the previous four steps, it is also possible to filter methods with the domain value of the framework. It does not mean that the method is only suitable for the listed domains, but it proves that the method has been successfully applied in those domains. The project resources decide the number of manageable candidate methods. In the end, the remaining methods are considered and will be implemented with the prepared dataset in the upcoming steps. The remaining sub-steps in this phase are the same as in general business or CRISP-DM projects. The candidate methods are implemented with the dataset prepared in the last phase, and their performance will be compared with the test design. Any method that reaches the test criteria can qualify for further evaluation.

Some methods, such as deep learning techniques, demand non-standard data preprocessing, such as an embedding of the used information. If (one of) the selected methods requires additional data preparation steps, it can be necessary to go back to the *Data Preparation* step and apply additional data preprocessing.

E. Evaluation

While the method assessment of the previous phase has a technical focus, the evaluation phase focusses on the business

Pred. type	Det. Pred. type	Input 1	Input 2	Input 3	Tool	Domain	Family of algorithms 1	Family of algorithms 2	Family of algorithms 3	Refer.
time	maint. time	event log (with timestamps)			N	automotive	time series	probabilistic model		[58]
time	activity delays	event log (with timestamps)			N	financial telecomm.	queueing theory	transition system		[61,62]
time	rem. time	event log (with timestamps)			N	telecomm.	stat. analysis			[6]
time	rem. time	event log (with timestamps)			ProM plugin	public admin.	transition system			[3,2]
time	rem. time	event log (with timestamps)			ProM plugin	customer supp.	stochastic Petri net			[55]
time	rem. time	event log (with timestamps)	process model		ProM plugin	financial customer supp.	regression	classification		[69]
time	rem. time	event log (with timestamps) with data			N	unspecified	pattern mining			[10]
time	rem. time	event log (with timestamps) with data			Y but unavail.	unspecified	classification	time series		[9]
time	rem. time	event log (with timestamps) with data			Y	financial public admin.	regression	classification		[68]
time	rem. time	event log (with timestamps) with data			Y	healthcare	stochastic Petri net			[60]
time	rem. time	event log (with timestamps) with data			ProM plugin	public admin.	regression			[20]
time	rem. time	event log (with timestamps) with data			ProM plugin	public admin.	transition system	regression	classification	[53]
time	rem. time	event log (with timestamps) with data			ProM plugin	customer supp. public admin. financial	transition system	regression		[54]
time	rem. time	event log (with timestamps) with data			ProM plugin	financial logistics	stochastic Petri net			[56,57]
time	rem. time	event log (with timestamps) with data	inter-case metrics		Y	healthcare manufacturing	regression			[59]
time	rem. time	event log (with timestamps) with data and contextual information			Y but unavail.	logistics	clustering	regression		[31]
time	rem. time	event log (with timestamps) with data and contextual information			Y	logistics	clustering	pattern mining		[5]
time	rem. time	event log (with timestamps) with data and contextual information			Y	logistics	clustering	transition system		[30]
time	rem. time	event log (with timestamps) with data and contextual information			ProM plugin	logistics	clustering	transition system		[29]
time	rem. time	event log (with timestamps) with data and contextual information	labeling funct.	proc. model	ProM plugin	no validation	classification			[38]

Fig. 7. Excerpt of a Framework of Existing PPM Methods [9]

perspective. The main criterion is whether the business goals can be reached by implementing the PPM method or not. If there is more than one method, which can achieve the business goals, the one performing best and being the most cost-efficient will be the most appropriate method for the project. The next step of the project is to deploy the method into the productive business environment. If there is no method satisfying the success criteria, the procedure model provides paths for this situation, which is tracing back to either *Method Selection and Implementation*, to apply alternative methods, or *Business Understanding*, if business goals can not be reached with the available methods.

F. Deployment

The *Deployment* phase in BPMP-PM is very similar to CRSIP-DM. What is specific for a PPM project is that the predictive quality should be continuously monitored in daily business. With constantly incoming instances, the prediction model should be updated regularly by being trained with new inputs. For this reason, monitoring and maintenance design should be considered in this step.

V. Pilot Application

As mentioned in the research approach, we used a pilot application to test the initial draft of the procedure model. The pilot used a real business case provided by the Business Process Intelligence Challenge 2019 [30] for demonstrating the application of the procedure model. The BPI Challenge 2019 provides the business case of an international company operating from the Netherlands, whose business is in the area of coating and paints. The provided real-life event logs stem from their daily business operations. The business case is about the purchase order handling process in the company. For the pilot application, we assume that the company paid a large amount of overdue fees in 2018. Therefore, the business goal of the PPM project is to reduce the overdue fee by foreseeing the payment date of each purchase order. The corresponding technical goal is to predict the remaining time of an ongoing purchase order from the current status to payment action. Due to size limitations and because the key contribution of this paper is the proposed procedure model and not the pilot application, we limit the description of the evaluation in the following to central information of the process. The dataset provided is the purchase order related event log submitted in 2018. The log contains about 1.5 million events, related to 76,349 purchase documents and 251,734 purchase items. The events are classified into 42 kinds of activities and have been executed by 627 users in the system. In this company, the activity “Clear Invoice” refers to paying the bill. Thus, we only selected the part of the process until “Clear Invoice” for further treatment in the *Data Preparation* phase. During *Select Candidate Methods*, the methods published by [31] (marked as [54] in Figure 7), [32] ([56] in Figure 7), and [33] ([20] in Figure 7) of the Time prediction type were the most suitable and the best-performing ones for this project. Unfortunately, a key piece of code of method [31] is currently not available, even though the Tool column is marked as “Y” (yes). This finding inspires us to update the framework in a future study. In the end, the remaining two methods have been implemented with the prepared dataset. In the technical evaluation, we compared the mean absolute error of each method. From a business perspective, an outcome evaluation means to compare the overdue costs that occurred in 2018 with the costs, which would have occurred if the predictive method were already in place. The deployment and maintenance costs should also be considered in this comparison. Finally, the method with the lowest overall cost is the most appropriate method for the project. In our case, the method [33] was the only one to achieve the business goals and hence selected for deployment.

During the pilot application, we have found that it often takes several iterations between Business Understanding and Data Understanding in order to determine the right technical goal. In the step Select Candidate Methods, the algorithm family turned out to be an efficient filter to eliminate inferior methods.

VI. Evaluation Concept

The pilot application, as well as our previous experiences with PPM implementations, support the design and content of the BPMP-PM. Nevertheless, appropriate formal evaluations are necessary to assess the performance regarding different perspectives, such as efficiency, failure rate, cost, benefit, and time consumption [34]. Therefore, we are currently performing two separate evaluations of the BPMP-PM, which we describe in detail in this section.

First, we are currently supporting an international producer of tabletop products, who wants to leverage PPM in their custom orders process. Together with them, we leverage the initial design of the BPMP-PM to guide their implementation process. Through workshops and individual interviews with the involved personnel, we continuously gather improvement potential and shortcomings of the procedure model, such that we can improve it in future iterations.

Second, we will apply case study research and evaluate the BPMP-PM against numerous different cases of PPM implementation. Case study research can be used to study an object in a contemporary real-life context [35]. It is especially useful when one wants to research contextual conditions. In such a condition, there usually exist many more variables than data points. In our case, we want to apply (practical) cases to the developed taxonomy (theory) to validate and improve it. Yin proposes an iterative 6-step approach for doing case study research. It comprises the steps plan, design, prepare, collect, analyze, share, and gives guidelines and principles for each step [35]. We will follow these steps and guidelines. A very crucial step at the beginning of case study research is a robust study design, and with it, the number of cases and their unit of analysis. In the case of PPM, the prediction goal plays a central role and will be considered as one dimension of analysis. Therefore, we will opt for a multiple case-based design, which improves external validity but also increases resource needs.

Unfortunately, freely available documentation of PPM endeavors is scarce to non-existent. Therefore, we will leverage the rich history of the Business Process Intelligence (BPI) Challenges, which are part of a regularly held workshop at the International Conference on Business Process Management. Each year a real-world event log data set is released in conjunction with the business case of a company. The business case formulates several research questions that are answered by researchers with their published papers. The data sets and business cases of the BPI Challenge series represent the defacto standard in data sets in the BPM and PPM domain, and researchers regularly benchmark their developments using these data sets. Therefore, we will investigate related publications of the BPI workshop solutions for BPMP-PM-relevant insights and use them as cases in our case study.

VII. Conclusion

This paper proposes a concept of an integrated procedure model (BPMP-PM) for implementing PPM methods in practice, starting from business background analysis until deployment. The model is generalized to fit all kinds of prediction goals, domains, and event logs of PPM. The model contains six steps and 19 sub-steps. The basis of the procedure model comes from CRISP-DM, which is the most commonly used process model for data mining projects. The steps towards successful completion of a business project are similar in data mining and PPM. However, most of the detail tasks in each step are designed differently because of the distinctions of the two fields. The differences are most significant in the *Data Understanding* and *Data Preparation* steps, since the central data type of PPM (event log) is not the same as in a generic data mining project. Additionally, the step *Method Selection and Implementation* was introduced to replace *Modeling*. It is designed to meet the specific needs of PPM projects. Our pilot application can only show a general feasibility, but does not evaluate the performance nor the overall validity of the model. In Section VI we described the next steps of our evaluation, which will enable us to further improve and validate the developed procedure model. The formal evaluation on a larger scale will allow us to measure the benefit that the application of this procedure model can bring to a business project [36].

Furthermore, we see a need to update the framework of existing PPM methods. On the one hand, new methods should be included into the framework and on the other hand, the comparison of method performance during method selection can help to clean out the irrelevant ones, so that only the most mature and available methods are included. Also, new trends of the PPM domain, such as including additional context information, should be considered in future developments of the framework.

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